Robert R. Ford

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EDUCATION

PhD, Climate Science, University at Albany, SUNY — Expected 2027
  Dissertation: *Mechanisms and impacts of multidecadal Southern Ocean convective variability*
  Advisor: B. E. J. Rose

BS, Applied Physics & Mathematics, Stockton University — 2021

RESEARCH EXPERIENCE

Graduate Research Assistant, University at Albany, SUNY — 2021 – Present

Undergraduate Researcher, Stockton University — 2019 – 2021

PUBLICATIONS

Submitted, in revision, etc.

1. Rose, B. E. J., **R. R. Ford**, A. Banihirwe, M. D. Camron, J. Clyne, O. Eroglu, K. FitzGerald, B. Freeman, M. A. Grover, J. Kent, R. May, K. Paul, K. R. Tyle, and A. Zacharias: Pythia Foundations: A community learning resource for Python-based computing in the geosciences. *In revision for Journal of Open Source Education*, https://jose.theoj.org/papers/6ad4fe1df16342a33f750b43772c1b15.

Peer-reviewed articles

2. **Ford, R. R.**, and B. E. J. Rose, 2026: A Southern Ocean Multidecadal Oscillator Forced by Deep Convection. *Geophysical Research Letters*, **53**, e2025GL120643, https://doi.org/10.1029/2025GL120643.

1. **Ford, R. R.**, B. E. J. Rose, and M. C. Rencurrel, 2025: Transient Climate Sensitivity Shaped by Low Cloud Changes Remotely Driven by Southern Ocean Processes. *Journal of Climate*, **38**, 797–813, https://doi.org/10.1175/JCLI-D-24-0164.1.

PRESENTATIONS

4. **Ford, R.**, and B. E. J. Rose, 2026: Exploring Mechanisms for Southern Ocean Convective Variability in a High-Resolution GCM [poster], Ocean Sciences Meeting.

3. **Ford, R. R.**, and B. E. J. Rose, 2025: The Role of Low-Frequency Variability and Open-Ocean Polynyas in Southern Ocean SST Trends [oral], AMS 18th Conference on Polar Meteorology and Oceanography.

2. **Ford, R.**, and B. E. J. Rose, 2023: Transient climate sensitivity shaped by Antarctic sea ice changes: exploring links between ocean heat uptake patterns, sea ice changes, and mid-latitude cloud cover [oral], AGU Fall Meeting.

1. **Ford, R.**, and J. R. Manson, 2021: Solute transport modeling using a Preissmann scheme [oral], 6th IAHR Europe Congress.

Coauthored presentations can be found at https://scholar.google.com/citations?user=PXTHuBAAAAAJ.

SOFTWARE

Developing Project Pythia Foundations and Cookbooks 2021 – Present

Source code can be found at https://github.com/r-ford and https://github.com/ProjectPythia.

TEACHING

University at Albany, SUNY

Instructor, AATM 103 Introduction to Climate Change	Summer 2026
Guest Lecturer, AATM 522 Climate Variability and Predictability	Spring 2026

Other

TA, Computational Tools for Climate Science, Climatematch Academy	Summer 2025
TA, PHYS 2110/2120 Physics for Life Sciences I/II, Stockton University	Fall 2019 – Spring 2021

HONORS & AWARDS

Paul Saraduke Jr. Memorial Physics Award, Stockton University	2021
Jason Shulman Award for Excellence in Physics Research, Stockton University	2020
Foundation Scholarship, Stockton University	2020

SERVICE & OUTREACH

Department of Atmospheric & Environmental Sciences, University at Albany, SUNY

Organizer, Climate Group seminar	2024 – 2026
Mentor, Undergraduate–graduate mentorship program	2024 – 2026
Tutor, Volunteer undergraduate tutoring (program lead 2024 – 2025)	2022 – 2025

Professional

Organizer, Project Pythia Cook-off hackathons held at NCAR	2023 – 2026
Contributed presentations on climate modeling to Climatematch Academy	2024

TECHNICAL SKILLS

Programming & tools: Python, Git, Bash, NCO, LaTeX, HPC environments
Models: CESM1/2

PROFESSIONAL AFFILIATIONS

American Geophysical Union
American Meteorological Society